

Lesson Exemplar in General Science

Quarter 3

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Lesson Exemplar for General Science
Quarter 3: Unit 4

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LESSON EXEMPLAR

Learning Area	GENERAL SCIENCE	Grade Level	11
Semester	SECOND	Quarter	3

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	<i>The learners learn that organ systems work together to carry out essential functions of an organism</i>
Performance Standard	<i>The learners use examples to describe the relationships among living systems and their components, as well as how the structure of a part relates to its function. They explain the role of cells in the perpetuation of life and how organ systems work together to maintain good health. They conduct investigations on the effects of climate change in local ecosystem.</i>
Learning Competencies	Key Topics Learned in Previous Grade Level • Observe the outer parts of plants, such as leaves, roots, and stems and identify their role to get water and nutrients from the soil. • Describe the purpose of specialized structures in plants, such as rhizomes, tubers, thorns, bulbs, and aerial roots. • Describe how plant organs (leaf, stem, roots) work together as the transport system In this lesson, Grade 11 learners will be able to: • Explain how various plant organs interact to facilitate the transport of materials throughout the plant system
II. REFERENCES and MATERIALS <i>(Selecting Resources and Material)</i>	• CK-12 Foundation. “9.8: Vascular Plants.” <i>Biology LibreTexts</i>. Accessed September 11, 2025. https://bio.libretexts.org/Bookshelves/Introductory_and_General_Biology/Introductory_Biology_(CK-12)/:A_Plants/9.08:_Vascular_Plants. • “Xylem and Phloem – Transport in Plants Biology.” <i>YouTube</i>. Video, 6:51. Posted May 30, 2021. https://www.youtube.com/watch?v=DhyYt1K844 • FuseSchool. “Xylem and Phloem – Transport in Plants Biology FuseSchool.” <i>YouTube</i>. Video, 4:30. Posted September 27, 2016. https://www.youtube.com/watch?v=jtuX7H05tmQ

III. CONTENT <i>(Sequencing Content)</i>	<i>Organ systems working together</i> • <i>Introduction to plant organ systems</i> • <i>Vascular tissues and transport mechanism</i> • <i>Coordinating among plant organs</i> • <i>Environmental influence and organ system responses</i>
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	<i>Unpacked Learning Competencies</i> 1. <i>Classify the major plant organs based on their roles in transporting water, minerals, and food.</i> 2. <i>Compare and contrast the structure and function of xylem and phloem, and assess how their design supports efficient material transport.</i> 3. <i>Analyze the impact of varying environmental conditions (e.g., drought, soil composition, temperature) on the interaction and efficiency of plant organs in material transports.</i> 4. <i>Construct a model or diagram that illustrates the transport pathways of water, minerals, and food through vascular tissues, and explain how each organ contributes to the process.</i>
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i>	ANNOTATION <i>*Instruction to the teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge	1. Activating Prior Knowledge <i>Before beginning the activity, the teacher introduces a simple yet thought-provoking idea: “Why do we need to water plants? When someone waters a plant, how does the water reach the leaves? Leaves produce foods, and when the leaves produce food, how is it transported to the rest of the plant organs? The teacher explains that although plants don’t have hearts or blood like humans, they possess their own system for moving water, minerals, and food.</i> <i>* To activate prior knowledge, the teacher encourages students to recall familiar plant parts and reflect on their functions. This sets the stage for identifying the key organs involved in transport and prepares learners for the day’s activity.</i> <i>* Teachers may select one or more of the suggested options to implement differentiated</i>

Option 1: Knowledge Quest

The teacher will ask the students to create three columns labeled: K – What do I KNOW? W – What do I WANT to know? H – How do I FIND OUT? The learners are encouraged to fill out all the three columns about household and personal care products

K	W	H
What I Know?	What I Want to know?	How do I find out?

Option 1: Think-Pair-Draw

Students will ask to list or sketch any plant organs they can recall, particularly those they believe are involved in transporting water, minerals, or food. The teacher instructs the students to pair up and share and discuss their ideas. They are tasked to draw a simple diagram of a plant, label the major organ and use arrows to show the movement of water, minerals and roots. Once the drawings are completed, they are to present their works to the class. The teacher facilitates a brief discussion highlighting correct identifications, clarifying misconceptions, and reinforcing the roles of each organ in the plant's transport system.

Note: Possible appropriate answer (sketch or write)

Table 1. Major Plant Organs Involved in Transport

Organ	Role in Transport
ROOTS	• Absorb water and minerals from the soil • Initiate upward transport through xylem • Anchor the plant and store nutrients
STEM	• Acts as a conduit for water, minerals, and food • Supports the plant and connects roots to leaves
LEAVES	• Site of photosynthesis (produces glucose) • Send food to other organs via phloem
FLOWERS	• Receive nutrients for reproduction
FRUITS	• Store sugars and nutrients

instruction based on their students' needs, learning styles, and classroom context. After the students complete the assigned activity, the teacher provides constructive feedback

* The IDF elements present in this section are reflective, in nature. It is reflective because the activity focuses on critical thinking, drawing from related experiences.

It is in the Analysis phase because it involves identifying learning goals, understanding learners' needs, analyzing tasks, and setting clear instructional objectives.

This is ideational because it emphasizes the things or characteristics that students see and experience (context) in order to perform the activity.

	Process Questions: 1. What plant parts do I already know? 2. Which organ do we think is most important for transport- and why? Option 2: Transport Trail Activity The teacher begins by distributing a simplified diagram of a plant, showing the root, stem, leaf, flower, and fruit. Arrows are included to represent the movement of water, minerals, and food, but the organs and pathways are left unlabeled. Students are instructed to work in pairs to label each organ and trace the correct path of water, minerals, and food. They are encouraged to use color coding or symbols to differentiate between the types of materials being transported. They are asked to present their completed Transport Trails. The teacher uses these presentations to clarify misconceptions, reinforce correct pathways, and highlight the coordinated roles of plant organs in transport. Note: Possible appropriate answer Table 2. Flow of Nutrients and Water Through Plant Organs <table><tr><th>Material</th><th>Entry point</th><th>Destination</th><th>Organs involved</th></tr><tr><td>Water</td><td>Roots</td><td>Leaves (for photosynthesis), other cells</td><td>Roots → Stem → Leaves</td></tr><tr><td>Minerals</td><td>Roots</td><td>Leaves, growing tissues</td><td>Roots → Stem → Leaves</td></tr><tr><td>Food (Glucose)</td><td>Leaves</td><td>Roots, stem, flowers, fruits</td><td>Leaves → Stem → Roots/Fruits/Flowers</td></tr></table> Process Questions:	Material	Entry point	Destination	Organs involved	Water	Roots	Leaves (for photosynthesis), other cells	Roots → Stem → Leaves	Minerals	Roots	Leaves, growing tissues	Roots → Stem → Leaves	Food (Glucose)	Leaves	Roots, stem, flowers, fruits	Leaves → Stem → Roots/Fruits/Flowers	The guide questions also serve as a formative assessment, allowing the teacher to evaluate the breadth and depth of students' understanding of the concept This belongs to the engage phase, where learners actively connect with the topic, explore ideas, and build curiosity before diving deeper into the lesson. * This section is both responsive and reflective as it recognizes the diverse backgrounds of students while emphasizing critical thinking and metacognition. It fosters an inclusive learning environment, encouraging students to analyze concepts deeply and connect their experiences to broader ideas. Design phase focuses on sequencing content, selecting appropriate strategies, and making learning meaningful which ensure coherence and engagement
Material	Entry point	Destination	Organs involved															
Water	Roots	Leaves (for photosynthesis), other cells	Roots → Stem → Leaves															
Minerals	Roots	Leaves, growing tissues	Roots → Stem → Leaves															
Food (Glucose)	Leaves	Roots, stem, flowers, fruits	Leaves → Stem → Roots/Fruits/Flowers															

	1. <i>In plants, which organ is primarily responsible for making their own food?</i> 2. <i>How does water, minerals and glucose reach or support the development of fruit or flower?</i> 2. Establishing the Purpose of the Lesson <i>The teacher will revisit the major organ of plants and its basic roles as introduced in the previous activity then pose a guiding thought: “Just like humans, plants also survive not because of one organ working alone, but because all parts work together as system</i> Option 1: Plant Emergency Scenario <i>Students are asked to reflect silently, then share initial thoughts with a partner. The teacher facilitates a brief discussion, prompting students to consider how different plant organs might be connected in solving the mystery.</i> Scenario: <i>A plant in the school garden has wilted leaves, shriveled fruits, and dry soil around its roots. The gardener wonders: What went wrong? Which plant parts might be failing, and how does that affect the rest of the plant?</i> <i>Process Questions:</i> 1. <i>What clues in the scenario suggest a problem?</i> 2. <i>Which organs might be involved in solving or causing the issue?</i> 3. <i>How might one organ’s failure affect the others?</i> Option 2: Organ Connection Web <i>Students are given cards or sticky notes labeled with different plant organs and materials (e.g., water, minerals, glucose). Students in</i>	<i>* Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>*The teacher asks the learners in advance to bring a large sheet of paper for this activity.</i>
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	groups arrange the cards on a large sheet of paper and draw arrows to show how each organ connects to the others in transporting materials. The teacher facilitates a brief sharing session, highlighting how each group visualized the system and reinforcing the idea that plant survival depends on organ coordination. Process Questions: 1. What happens if one organ doesn't perform its function properly? 2. Why is it important for all organs to work together as a system?	
B. Instituting New Knowledge	1. Presenting Examples The teacher briefly recalls the previous lesson, where students explored how plant organs work together to transport water, minerals, and food. To spark curiosity, the teacher poses a guiding question: "If we think of the stem as a busy road and your destination is the leaves, what acts as the vehicle that carries water from the roots up to the leaves?" Option 1: Slides Presentation The teacher will facilitate guided discussion using slide presentation and process questions. Concepts to discuss • Vascular plants are known as tracheophytes, which literally means "tube plants." • Vascular tissues are specialized to transport fluid. They consist of long, narrow cells arranged end-to-end, forming tubes. There are two different types of vascular tissues, called xylem and phloem 1. Xylem is vascular tissue that transports water and dissolved minerals from roots to stems and leaves. This type of tissue consists of dead cells that lack end walls between adjacent cells. The side walls are thick and reinforced with lignin, which makes them stiff and water proof. 2. Phloem is vascular tissue that transports food (sugar dissolved in water) from photosynthetic cells to other parts of	* This remains a formative assessment, designed to provide insight into students' understanding. It is integrative, as it takes a broad perspective on concepts, considering various aspects. It is also linked to context and connection, focusing on students' backgrounds and related experiences. Furthermore, it falls under the explore phase, as the activity allows students to navigate and deepen their knowledge. * The teacher will choose one activity from the given options This approach is inclusive, ideational, and integrative. The processing questions acknowledge various contexts, facilitate conceptual understanding, and provide a holistic perspective. It also emphasizes context

the plant for growth or storage. This type of tissue consists of living cells that are separated by end walls with tiny perforations, or holes.

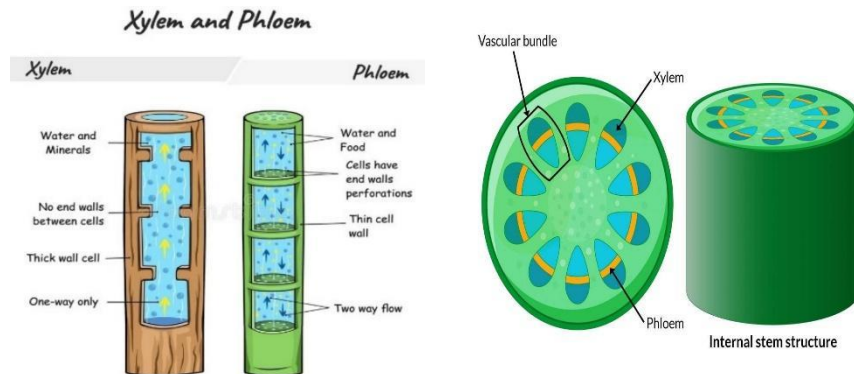


Figure 1. Xylem and Phloem/GCSE Biology Revision

Table 3. Xylem and Phloem: The Transport Tissues

Tissue	Main Function	Direction of Flow	Materials Moved	Connected Organs
Xylem	Transports water and minerals from the roots upward	One-way (roots → stem → leaves)	Water, dissolved minerals	Roots, stem, leaves
Phloem	Transports food (glucose) made in the leaves to other parts	Bidirectional (source → sink)	Sugars, nutrients	Leaves, stem, roots, fruits, flowers

How They Interact

- Water and minerals: Move from roots → stem → leaves via xylem

and connection, ensuring that learners relate their experiences to the discussion.

- Food (glucose): Moves from leaves → stem → roots/fruits/flowers via phloem

COMPONENTS OF VASCULAR TISSUE

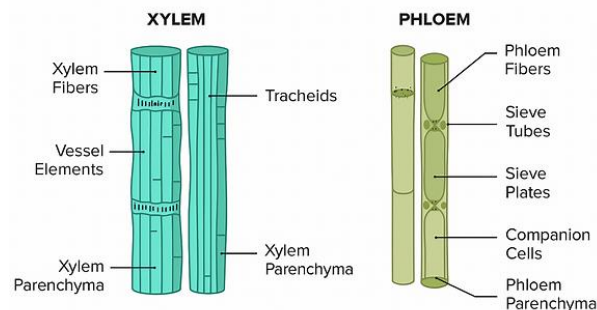


Figure 2. Components of Phloem Vascular tissue

Process Questions:

1. How does the structure of xylem and phloem support their specific roles in transporting water and food?
2. What might happen if one tissue type were missing?"

Table 4. Xylem Tissue Components and Their Functions

Component	Function
Vessel Elements	Wide, hollow cells stacked end-to-end; conduct water rapidly through perforations.
Tracheid	Long, tapered cells with pits; conduct water more slowly and provide support.
Xylem Parenchyma	Living cells; store nutrients and assist in lateral transport of water.
Xylem Fibers	Thick-walled supportive cells; provide mechanical strength to the plant.
Lignin (<i>not a cell but a key substance</i>)	Strengthens cell walls and waterproofs xylem vessels.

* Students will be guided to answer the process questions. The teacher facilitates the activity and provides constructive feedback.

Table 5. Phloem Tissue Components and Their Functions

Component	Function
Sieve Tube Elements	Main conducting cells; transport sugars and other organic materials.
Companion Cells	Regulate sieve tube activity; assist in loading and unloading of sugars.
Phloem Parenchyma	Storage and lateral transport of food substances.
Phloem Fibers	Provide structural support; often found in bundles.
Sieve Plates (<i>part of sieve tubes</i>)	Perforated end walls that allow flow between sieve tube cells.

Option 2: Mind Mapping the Vascular Highway

Students are guided to place the central concept “Vascular Bundles in Plants” at the center of their page. From there, they branch out into two main arms: Xylem and Phloem. Under each, they add sub-branches that include:

- *Tissue components (e.g., vessel elements, sieve tubes, companion cells)*
- *Functions (e.g., water transport, food transport, support)*
- *Direction of flow (e.g., upward for xylem, bidirectional for phloem)*
- *Associated plant organs (e.g., roots, stems, leaves)*

The teacher encourages students to use color coding, arrows, symbols, and analogies to make their maps meaningful.

2. Discussing New Concept

The teacher begins with a formative assessment before proceeding the new concept by answering five items TRUE or FALSE

**This approach is relevant and responsive, incorporating transdisciplinary learning and problem-solving. It transcends language and the formation of ideas, connecting with various fields of knowledge. Additionally, it is solution-oriented, focusing on addressing challenges.*

Furthermore, because it nurtures multiple skills and areas of expertise, it is integrative—encouraging learners to apply diverse knowledge. The process also fosters collaboration and creativity.

	1. <i>Xylem transports water and minerals from the roots to the leaves.</i> 2. <i>Phloem transport is bidirectional, while xylem transport is unidirectional.</i> 3. <i>Xylem cells are dead at maturity, while phloem cells are alive.</i> 4. <i>Phloem contains sieve tube elements and companion cells.</i> 5. <i>Xylem vessels have thick lignified walls to provide structural support.</i> <i>Great job thinking through those statements! Now, let's bring those ideas to life. Picture this: a mango tree basking in sunlight, its roots deep in the soil, leaves working hard to make food, and fruits growing sweet and full. But here's the puzzle—how do water and minerals from the roots reach the leaves, and how do sugars from the leaves travel all the way to the fruits at the top? Let's explore the incredible transport system inside plants that makes this possible</i> Option 1. Video lesson • Xylem and Phloem - Transport in Plants Offers a clear breakdown of vascular tissue arrangement in roots, stems, and leaves. It's especially helpful for connecting structure to function across different plant organs. (2021, May 30). <i>Xylem and Phloem - Transport in Plants Biology ...</i> [Video]. YouTube. https://www.youtube.com/watch?v=DhyYtT1K844 • Xylem and Phloem - Transport in Plants Biology FuseSchool A concise and animated explanation of vascular bundles, xylem structure, and phloem function. Great for visual learners and for reinforcing the idea of directionality in transport. FuseSchool. (2016, September 27). <i>Xylem and Phloem – Transport in Plants Biology FuseSchool</i> [Video]. YouTube. https://www.youtube.com/watch?v=jtuX7H05tmQ	<i>*Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i>
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	Process Questions: 1. What materials are transported by each tissue, and in which direction? 2. How do the structures of xylem and phloem support their transport roles? 3. What evidence supports the idea that vascular tissues are connected all plant organs? 4. Which part of the vascular system do you think is more vulnerable to environmental stress? Why? Option 2. Flow Forces: Modeling Transport in Plants Students will explain the mechanisms of water and food transport in xylem and phloem, and analyze why their flow directions differ, using models and guided reasoning. Part 1: Xylem Transport Model Create a visual or physical model that shows how water moves upward through the xylem. Your model should include: • Transpiration pulls • Cohesion and adhesion • Capillary action • Unidirectional flow Part 2: Phloem Transport Simulation Design a model or diagram that explains how food (sugars) moves through the phloem using the pressure-flow hypothesis. Your model should show: Source and sink dynamics Bidirectional flow Active transport and pressure gradients Process Questions 1. What forces drive water upward through xylem (e.g., transpiration pull, cohesion-tension)? 2. How does the pressure-flow hypothesis explain food transport in phloem?	
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3. Why is phloem transport bidirectional while xylem is unidirectional?

Concepts to discuss

- **Transpiration pull** is the upward force that draws water from the roots to the leaves through the xylem. It happens when water evaporates from leaf surfaces (mainly through stomata), creating a negative pressure that pulls more water upward.
- **Cohesion** is the attraction between water molecules (they stick to each other).
- **Adhesion** is the attraction between water molecules and the walls of xylem vessels.
- **Capillary action** is the ability of water to move upward through narrow tubes (like xylem vessels) due to cohesion and adhesion.
- **Unidirectional flow** means movement in only one direction. In plants, water and minerals move **upward** through the xylem—from roots to leaves.
- In phloem transport, the **source** is the part of the plant where sugars (mainly sucrose) are produced—usually the leaves through photosynthesis. The **sink** is where those sugars are needed or stored—such as roots, fruits, or growing shoots.
- Bidirectional flow means movement is in both directions—upward and downward. In plants food can be transported in both directions through phloem transport food.

3. Developing Mastery

At this stage of the lesson, the teacher guides students to move beyond basic identification of vascular tissues and their functions. The teacher introduces the next task by framing the plant as a coordinated system:

“In every living plant, the roots, stem, and leaves do not work in isolation. They interact constantly—absorbing, producing, and

	<i>distributing materials to keep the plant alive and growing. Just like a well-organized team, each organ has a role, and the vascular tissues act as the communication and delivery lines between them.</i> <i>Students are reminded that mastery of the topic involves recognizing not just what each part does, but how they work together</i> Option 1: Role Play Activity: “Inside the Plant Transport System” <i>The students dramatize how roots, stems, and leaves coordinate through xylem and phloem to transport water, minerals, and food throughout the plant</i> <i>The teacher assigns students into small groups and introduces the role play activity. Each group is tasked with dramatizing how water, minerals, and food are transported throughout the plant, emphasizing the interaction among roots, stems, leaves, and vascular tissues.</i> <i>Each student in the group is assigned a specific role, such as:</i> ● Root – responsible for absorbing water and minerals from the soil ● Stem – serves as the transport highway connecting roots and leaves ● Leaf – produces food through photosynthesis and regulates transpiration ● Xylem – transports water and minerals upward from the roots ● Phloem – distributes sugars from the leaves to other parts of the plant <i>Optional roles may include Stomata, Sunlight, or Sink Organs (e.g., fruits or root tips)</i> <i>Props and Visuals (Optional):</i> ● <i>Blue ribbons or ropes for xylem flow</i> ● <i>Green ribbons for phloem flow</i> ● <i>Signs or headbands with role names</i> ● <i>Arrows or labels showing direction of transport</i> <i>Scene 1 Water Uptake</i> <i>Scene 2: Transpiration Pull</i>	<i>*Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i>
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Scene 3: Food Production and Distribution
Scene 4: System Coordination

Table 6. Rubric Role Play Activity: “Inside the Plant Transport System

Criteria	Excellent (5 pts)	Good (3 pts)	Needs Improvement (1 pt)
Content Accuracy	Shows clear and correct understanding of plant transport.	Mostly correct with minor errors.	Many mistakes or unclear concepts.
Participation	Everyone is actively involved and plays their role well.	Most students participate.	Few students participate or roles are unclear.
Creativity	Uses props, movement, or dialogue creatively.	Some creative elements used.	Little or no creativity shown.
Clarity of Presentation	Easy to follow and understand.	Mostly clear.	Hard to follow or confusing.

Options 2: Plant Transport System: Organ Interaction Map

Students are sked to visualize how these tissues operate across different organs—roots, stems, and leaves—and how these organs work together as part of a unified transport system by creating a labeled diagram that includes:

- The **three major plant organs**: roots, stem, and leaves
- The **vascular tissues** (xylem and phloem) within each organ
- **Arrows and color coding** to show the direction of water and food movement
- **Short annotations** explaining each organ’s contribution to the transport process

**While suggested rubrics are available to guide assessment of student understanding during this activity, the teacher has opted not to use them for this segment.*

	Process Questions: 1. How do the roots, stem, and leaves coordinate to ensure continuous transport of water and nutrients throughout the plant? 2. How does the interaction among organs reflect the idea of the plant as a unified system rather than separate parts?	
C. Demonstrating Knowledge and Skills	1. Finding Practical Application <i>The teacher guides students to connect their understanding of vascular tissues and organ interaction to real-world contexts.</i> Option 1: Plant Transport in Action: Solving Real-World Challenges <i>The teacher presents students with realistic scenarios where the plant transport system is disrupted or challenged. Students work in small groups to analyze the situation, identify which part of the transport system is affected, and propose science-based solutions or interventions.</i> <i>Each group receives one of the following scenarios:</i> • A farmer notices that crops are wilting despite regular watering. • A fruit tree produces small, underdeveloped fruits. • A plant exposed to drought shows signs of leaf curling and stunted growth. • A student over-fertilizes a potted plant and sees leaf burn and root damage. Option 2: Investigating Climate Change, its Impact on Plant Transport Systems? <i>Students will conduct a research-based investigation to understand how climate change affects the transport systems in plants, specifically focusing on xylem and phloem functions.</i>	<i>*Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>This section is both relevant and reflective, as it hones creativity, critical thinking, and analytical skills. It encourages learners to engage deeply with concepts, fostering a thoughtful and insightful approach to understanding.</i>

To demonstrate their understanding, students will present their findings in 500–800 words

Each output should include:

- A clear explanation of xylem and phloem functions
- The effects of climate-related stress (e.g., drought, heat) on plant transport
- Visuals or diagrams to support their explanation
- At least two credible sources (e.g., scientific articles, climate reports)

Table 5. Rubric Role Play Activity: “Inside the Plant Transport System

Criteria	Excellent (5 pts)	Good (3 pts)	Needs Improvement (1 pt)
Understanding of Topic	Demonstrates deep understanding of climate change effects on xylem and phloem.	Shows clear understanding with minor gaps.	Limited understanding; major misconceptions
Visual and Presentation	Includes relevant visuals (diagrams, charts) that enhance understanding	Visuals are present but may be basic or loosely connected.	No visuals included.
Research Sources	Uses 2+ credible sources effectively and integrates them into the explanation.	Uses sources but with limited integration or explanation.	No sources used or sources are unreliable.

2. Making Generalization

The teacher guides students to consolidate their understanding of how various plant organs interact to facilitate the transport of materials. Students are prompted to articulate a general principle or summary statement that captures the essential idea of the lesson—how organ-

**This is reflective, as it highlights critical thinking and metacognition in the development of the activity and its explanation. Additionally, the empathize stage is present, where students can express their ideas and connect personally with the subject matter.*

While suggested rubrics are available to guide assessment of student understanding during this activity, the teacher has opted not to use them for this segment.

match each organ or tissue to its correct function and then choose one pair to explain in 1–2 sentences.

Column A (Structure)	Column B (Function)
1. Roots	A. Absorb water and minerals from the soil and anchor the plant
2. Xylem	B. Transport water and minerals from roots to leaves
3. Phloem	C. Transport sugars from leaves to other parts of the plant
4. Companion cell	D. Main phloem cells that conduct sugars throughout the plant
5. Tracheids	E. Assist sieve tubes in loading and unloading sugars
6. Sieve tubes	F. Conduct water and provide structural support

4. Additional Activities

To reinforce conceptual understanding and allow for creative expression, the teacher asks the students to do an individual performance task.

Option 1: Visualizing the Flow: Plant Transport Infographic

Students will create an infographic that illustrates and explains the transport of water, minerals, and food in plants, highlighting the roles of roots, stem, leaves, xylem, and phloem

Option 2 Learning Summary Report

The teacher assigns an individual written task titled “Learning Summary Report.” This activity allows students to organize their understanding into a structured outline that highlights the essential concepts covered in the lesson. Each student is instructed to complete a report outline with the following sections:

Title: Understanding the Plant Transport System: Organ Interaction and Vascular Function

**Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context.*

	• Introduction • Key Concepts learned • Systems Thinking Insight • Real-World Connection • Generalization Statement Option 3. Knowledge & Next Steps The teacher asks the learners organize their responses into a structured table with three columns <table><tr><th>L</th><th>A</th><th>Q</th></tr><tr><td>What I have learned?</td><td>What action will I take?</td><td>What new questions do I have?</td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	L	A	Q	What I have learned?	What action will I take?	What new questions do I have?							Use the KWH activity in the beginning of the lesson if you will utilize the LAQ activity
L	A	Q												
What I have learned?	What action will I take?	What new questions do I have?												
V. ASSESSMENT (Assessing Learnings)	The teacher instructed the students to read each question carefully and select the best answer based on what they have learned. Directions. Read each item carefully and write the letter of the correct answer in a separate sheet of paper. - Which plant tissue is responsible for transporting water from the roots to the leaves? A. Cambium B. Phloem C. Epidermis D. Xylem - Why does damage to the xylem tissue in a plant affect the movement of water to the leaves? A. Because xylem stores food for the plant B. Because xylem helps in photosynthesis in the leaves C. Because xylem controls the opening and closing of stomata D. Because xylem transport water and minerals from roots to leaves - If the phloem in a plant is blocked, what would most likely happen to the roots? A They would receive more water C. They would begin photosynthesis B. They would stop absorbing minerals D. They would not receive enough sugars for growth - Why is the stem important in the transport of materials within a plant? A. It performs photosynthesis C. It absorbs water from the soil B. It stores food for the plants. D. Connects roots and leaves, allowing transport through vascular tissues													

	5. How do xylem and phloem differ in their transport roles? A. Xylem and phloem both transport water B. Xylem and phloem both transport food C. Xylem transport food: phloem transport water D. Xylem transports water and minerals: phloem transports sugars and nutrients 6. What happens if the phloem in a plant is damaged? A. Water cannot reach the leaves C. Sugar cannot be transported to growing parts B. Photosynthesis stops completely. D. Minerals and water cannot be absorbed from the soil 7. Which component of phloem assists in loading and unloading sugars into sieve tubes? A. Companion cells B. Tracheids C. Vessel elements D. Xylem fibers 8. Which xylem component helps conduct water and also provides structural support to the plant? A. Companion cells B. Phloem fibers C. Sieve tube elements D. Tracheids 9. A student claims that the phloem is more important than the xylem because it transports food. How should this claim to be evaluated? A. Agree, because food is more essential than water B. Agree, because xylem only supports the plant structurally C. Disagree, because phloem does not transport anything essential D. Disagree, because xylem also plays a vital role in transporting water and minerals. 10. Which of the following best evaluates the importance of organ interaction in plant transport? A. The stem alone manages all transport through vascular tissues B. Transport occurs independently in each organ without coordination C. Only the leaves are responsible for transport since they produce food D. Roots, stem, and leaves must work together to ensure efficient transport of water, minerals, and food
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>

APPENDIX

Activity 1: Flow Forces: Modeling Transport in Plants

Objective: Explain how water, minerals, and food are transported through the xylem and phloem, and analyze the roles of plant organs in facilitating these processes

Part 1: Xylem Transport Model

Create a visual or physical model that shows how water moves upward through the xylem. Your model should include:

- Transpiration pulls
- Cohesion and adhesion
- Capillary action
- Unidirectional flow

Part 2: Phloem Transport Simulation

Design a model or diagram that explains how food (sugars) moves through the phloem using the pressure-flow hypothesis. Your model should show:

- Source and sink dynamics
- Bidirectional flow
- Active transport and pressure gradients

Optional Extension

Add color coding, arrows, and analogies to your models. For example:

- Blue arrows for water flow
- Green arrows for sugar transport
- Straws for xylem, delivery trucks for phloem

Answer the following process questions

1. What forces drive water upward through xylem, and how do they work together
2. How does the pressure-flow hypothesis explain food transport in phloem?
3. Why is phloem transport bidirectional while xylem is unidirectional?

Assessment Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Work (1)
Accuracy of Explanation	All mechanisms clearly explained	Most mechanisms explained	Some mechanisms explained	Major misconceptions present
Clarity of Model/Diagram	Highly organized and easy to interpret	Mostly clear	Some confusion	Unclear or incomplete
Depth of Reasoning	Insightful answers with justification	Reasonable answers	Basic responses	Minimal effort
Creativity & Use of Analogies	Creative and relevant analogies used	Some creative elements	Limited creativity	No analogies or visuals

Team Collaboration	All members contributed actively	Most members involved	Uneven participation	Minimal collaboration
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True or False with Corrected Statements

- Water and minerals absorbed by the roots are transported to the leaves through the phloem. **FALSE**
Corrected Statement: Water and minerals absorbed by the roots are transported to the leaves through the xylem.
 - The stem contains vascular tissues that connect the roots and leaves, allowing continuous transport. **TRUE**
 - Phloem transports food produced in the leaves to all parts of the plant, including the roots. **TRUE**
 - Xylem transports water in both upward and downward directions depending on the plant's needs. **FALSE**
Corrected Statement: Xylem transports water only upward from the roots to the leaves.
 - The interaction among roots, stem, and leaves ensures that materials are distributed efficiently throughout the plant. **TRUE**
- Match and Explain

Column A (Structure)	Column B (Function)	Answer
1. Roots	A. Absorb water and minerals from the soil and anchor the plant	A
2. Xylem	B. Transport water and minerals from roots to leaves	B
3. Phloem	C. Transport sugars from leaves to other parts of the plant	C
4. Companion cell	D. Main phloem cells that conduct sugars throughout the plant	E
5. Tracheid	E. Assist sieve tubes in loading and unloading sugars	F
6. Sieve tubes	F. Conduct water and provide structural support	D

Explanations:

- Roots → A Roots absorb water and minerals from the soil and anchor the plant. They are the entry point for materials that xylem will later transport.
- Xylem → B Xylem is responsible for transporting water and dissolved minerals from the roots upward to the leaves. It consists of vessel elements and tracheids.
- Phloem → C Phloem transports sugars (products of photosynthesis) from the leaves to other parts of the plant, including growing tissues and storage organs.
- Companion cell → D Companion cells are closely associated with sieve tubes and help load and unload sugars into the phloem, ensuring efficient transport.
- Tracheid → E Tracheids are elongated xylem cells that conduct water and also provide structural support due to their thick lignified walls.
- Sieve tubes → F Sieve tube elements are the main conducting cells of the phloem. They form continuous channels for sugar transport but rely on companion cells for metabolic support.

ASSESSMENT

- Correct Answer: D. Xylem

- Explanation: Xylem is the vascular tissue that conducts water and dissolved minerals upward from the roots to the leaves.*
2. *Correct Answer: D. Because xylem transports water and minerals from roots to leaves*
Explanation: Xylem is essential for upward water transport. Damage disrupts this flow, leading to dehydration in the leaves.
3. *Correct Answer: D. They would not receive enough sugars for growth*
Explanation: Phloem transports sugars from the leaves to other parts of the plant, including the roots. A blockage prevents energy distribution, affecting root development.
4. *Correct Answer: D. Connects roots and leaves, allowing transport through vascular tissues*
Explanation: The stem houses xylem and phloem, forming a conduit between roots and leaves for material transport.
5. *Correct Answer: D. Xylem transports water and minerals; phloem transports sugars and nutrients*
Explanation: Xylem moves water upward; phloem distributes sugars from photosynthesis throughout the plant.
6. *Correct Answer: C. Sugar cannot be transported to growing parts*
Explanation: Phloem carries sugars to roots and buds. Damage prevents energy distribution, affecting growth.
7. *Correct Answer: A. Companion cells*
Explanation: Companion cells regulate sugar movement into and out of sieve tube elements for efficient transport.
8. *Correct Answer: D. Tracheids*
Explanation: Tracheids are elongated xylem cells that conduct water and reinforce the plant with lignified walls.
9. *Correct Answer: D. Disagree, because xylem also plays a vital role in transporting water and minerals*
Explanation: Both tissues are essential—xylem supports hydration and nutrient flow, which are critical for photosynthesis and survival.
10. *Correct Answer: D. Roots, stem, and leaves must work together to ensure efficient transport of water, minerals, and food*
Explanation: Transport is a coordinated system—roots absorb, stems conduct, and leaves produce food, all working in synergy.
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